

MATAN Z. YABLON

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EXPERIENCE

UROP, Engineering Quantum Systems Group at MIT (March 2025 – Present)

- Experimented with filtering, thermalization and attenuation techniques to optimize setup.
- Modeled attenuation in coaxial cables and antenna modes in superconducting qubits via HFSS and NumPy to control thermal noise sources.
- Developed multiplexed qubit readout capability in Python and QUA framework on OPX Plus. Work supported by Ralph L. Evans (1948) Endowment Fund and John Reed Fund.

Kupcinet-Getz Summer Scholar, Weizmann Institute of Science (June 2026 – August 2026)

- Spearheaded experimental research project utilizing superconducting transmon qubits to probe fundamental dynamics in quantum measurement.
- Engineered predictive Python simulations for experimental modeling, processing the resulting hardware data to detect and quantify theoretical quantum phenomena.

Student Intern, Thomas Jefferson National Accelerator Laboratory (June 2024 – August 2024)

- Conducted electronics testing on APV25 chips and GEM detectors by measuring cosmic signals.
- Detectors were to be used in the Large Area Detection (LAD) experiment, to understand the internal structure of the nucleus and EMC effect.

Participant, MIT PRIMES (2022 – 2023)

- Conducted technical research in machine learning involving strategies to make reinforcement learning algorithms robust. Presented research at MIT in October 2022 and May 2023.
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EDUCATION

Massachusetts Institute of Technology, Cambridge, MA (Aug 2023 – May 2027)

- 5.0/5.0. Pursuing Bachelor of Science in Computer Science (6-3) and Physics (8). Relevant coursework includes 8.06 Quantum Mechanics III, 8.701 Introduction to Particle Physics, 8.033 Relativity I, 8.09 Classical Mechanics III, 6.122 Design and Analysis of Algorithms, 18.404 Theory of Computation, 6.191 Computation Structures, 6.102 Software Construction

Harvard University Extension School (2021 – 2023)

- Four semesters of Ancient Greek, Beginning Ancient Greek I & II, Intermediate Ancient Greek I & II
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SKILLS

- Python, QUA, NumPy/SciPy, LaTeX, Java, C, TypeScript
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HONORS & AWARDS

- Qualified for AIME (2022)
 - USA Computing Olympiad (USACO) Silver Rank (2020)
 - MIT EECS | Analog Devices Undergraduate Research and Innovation Scholar (2025)
 - National Merit Scholarship Finalist (2023)
 - National Latin Exam Gold Medal, Summa Cum Laude (2022)
 - Massachusetts State Seal of Biliteracy in Latin (with Distinction) and Hebrew (2022)
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ADDITIONAL

- *Languages:* Hebrew, Latin, Ancient Greek, Arabic, Mandarin